

Trigonometric Functions & the Unit Circle



Radians, degrees, and the unit circle

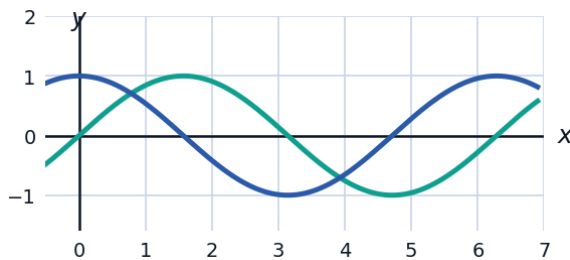
- 1 Convert 150° to radians, and convert $\frac{5\pi}{4}$ radians to degrees.
 - 2 Find the exact coordinates of the point on the unit circle at angle $\theta = \frac{2\pi}{3}$.
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Exact values from the unit circle

- 3 Find the exact values of $\sin\left(\frac{7\pi}{6}\right)$ and $\cos\left(\frac{7\pi}{6}\right)$.
 - 4 Find the exact value of $\tan\left(\frac{5\pi}{3}\right)$.
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Visualizing sine and cosine graphs

- 5 The graph shows $f(x) = \sin(x)$ and $g(x) = \cos(x)$ over one period.



- a State the amplitude and period of each function shown.
 - b Identify the horizontal shift that maps $\sin(x)$ onto $\cos(x)$.
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Amplitude, period, and vertical shift

- 6 State the amplitude, period, and vertical shift of $f(x) = 3\sin(2x) + 1$.
 - 7 A Ferris wheel's height above the ground is modeled by $h(t) = 20\cos\left(\frac{\pi}{6}t\right) + 22$, where t is in minutes. Find the maximum height, minimum height, and period of one full rotation.
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Reciprocal and Pythagorean identities

- 8 Given $\sin \theta = \frac{3}{5}$ with θ in Quadrant II, find $\cos \theta$ and $\tan \theta$.

- 9 Find $\sec \theta$ and $\csc \theta$ if $\cos \theta = -\frac{5}{13}$ and θ is in Quadrant III.
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Modeling periodic phenomena

- 10 The average daily temperature in a city is modeled by $T(m) = 15\sin\left(\frac{\pi}{6}(m - 3)\right) + 18$ degrees Celsius, where m is the month number ($m = 1$ for January). Find the temperature in April ($m = 4$) and state the month of maximum temperature.
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Co-function and even/odd identities

- 11 Use a co-function identity to rewrite $\cos\left(\frac{\pi}{5}\right)$ in terms of sine.
- 12 Use the fact that sine is an odd function and cosine is an even function to evaluate $\sin(-30^\circ)$ and $\cos(-60^\circ)$.
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Solving basic trigonometric equations

- 13 Solve $2\sin \theta - 1 = 0$ for θ in $[0, 2\pi)$.
- 14 Solve $2\cos^2 \theta - 1 = 0$ for θ in $[0, 2\pi)$.
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Angular and linear speed

- 15 A wheel of radius 10 cm rotates at 4 revolutions per second. Find its angular speed ω in radians per second, and the linear speed of a point on its rim.
- 16 A satellite orbits Earth at a radius of 7000 km, completing one orbit every 2 hours. Find its linear speed in km/h.
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Right-triangle definitions and applications

- 17 A ladder leans against a wall, making a 65° angle with the ground. If the ladder is 12 ft long, find the height it reaches on the wall, to one decimal place.
- 18 From a point 50 ft from the base of a tower, the angle of elevation to the top is 40° . Find the height of the tower, to one decimal place.

- 19** A kite string is 80 ft long and makes a 55° angle with the ground. Find the height of the kite above the ground, and the horizontal distance from the flyer to a point directly below the kite, both to one decimal place.